IAE PARIS — SORBONNE BUSINESS SCHOOL · IDGP ACADEMY

Digital Tools & Benchmarking

Reading notes for the afternoon session. Module 3 — Methods, Tools and Operational Efficiency, Session 4.

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No preparation is required. The afternoon tests the morning's argument against the two things organisations buy when they want capability and cannot build it: technology, and other people's practices.

Contents

What a digital investment actually buys; what an AI system was never told and what nobody measured; why best practice resists transfer; and how a transformation is run when the path cannot be known in advance.

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> **The instrument.** The group activity in Part Seven is run on a separate booklet, the *Digital Tool Appraisal*, which is fillable online and printed here as a companion. Bring one tool you are evaluating, or one you have already bought and are not sure about.

This morning ended on a claim: value is not the sum of what an organisation owns and how it works, but that sum scaled by its capacity to improve. The afternoon tests the claim against the two things organisations buy when they want that capacity and cannot build it — technology, and other people's practices.

The order is deliberate. Digital platforms come first because they are the most expensive thing on the list and the easiest to mistake for capability. Artificial intelligence comes next because it is the extreme case of the same error. Benchmarking comes last because it is the only one that looks like it should work.

> **WHAT TO BRING FROM THE MORNING**
>
> **Your Value Case canvas** — the project your group appraised. This afternoon examines the tooling that project depends on.
>
> **Your NUMMI hypothesis**, and the resolution we reached. The same mechanism explains why buying enterprise software does not build competitive advantage, and the puzzle below is its digital twin.

ONE

The digital puzzle

Two firms, one platform

The setting. Two organisations in the same industry, of comparable size, buy the same enterprise platform from the same vendor in the same year. Same modules, same implementation partner, comparable budgets. One of them also deploys the same demand-forecasting model, trained on a similar history.

Three years later, one has materially shorter lead times and a forecast its planners trust. The other has the platform, the licences, the dashboards — and a planning process that works around all three.

Nothing in the purchase differed. **Where did the difference come from, and why was it invisible in both business cases?**

If this sounds like Fremont with better software, that is the point. The morning established that assets and processes are additive and purchasable while capability is a coefficient. This afternoon the assets are more expensive, the vendors more persuasive, and the arithmetic unchanged.

TWO

What a digital investment actually buys

Almost every platform is an **A** decision. Some are also a **P** decision. Very few are a **C** decision — and the error that destroys the business case is not buying A, it is booking the benefit as though C had moved.

$$V = C \times (A + P)$$

a platform enters the bracket · the coefficient is bought separately, or not at all

The landscape, in six categories

The market is large, it changes faster than any syllabus, and nothing below is a recommendation. The categories are what matter, because **the category tells you what a tool does and the VCAP term tells you what it is worth**. The named products are examples only, offered so the categories are recognisable.

Category	Examples	What it typically changes	Usual term
Collaboration and work management	Asana, Monday, Trello, Planner, Jira	Visibility of tasks and ownership	A, and P if the flow of work is genuinely redesigned
Scheduling and portfolio	MS Project, Primavera P6, Planview, Clarity	Baselines, dependencies, resource and portfolio views	A, occasionally P
Analytics and dashboards	Power BI, Tableau, Looker	What leadership can see, and how often	A — and see the caution below
Integration and the data layer	APIs, integration platforms, a warehouse or lakehouse	Where the authoritative version of each fact lives	Frequently the only genuine P in a programme
Automation and workflow	Workflow engines, robotic process automation	Steps performed without a person	P, and A
AI and assistive	Assistants, forecasting models, document generation, agents	Predictions, drafts, and increasingly actions	A — Parts Three and Four are about why

Two observations that survive every change in the market. The least visible category — integration and the data layer — is the one that most often carries the real process change, and it is the one hardest to put in a business case because it demonstrates nothing on a screen. And the most visible category, dashboards, changes what is seen without changing what is done, unless somebody also changes the decision the dashboard feeds.

> **A CAUTION ON DASHBOARDS, CARRIED OVER FROM THIS MORNING**
>
> A dashboard reports variation; it does not distinguish the two kinds. A governance routine that demands an explanation for every movement against target — favourable or adverse — is treating common-cause variation as though it were special cause, which this morning established makes the process worse rather than better. The instrument is not the problem. The routine attached to it is, and the routine is a **P** decision that almost never appears in the tooling business case.

The triage, applied to a tool

Term	What a tool contributes	The test
A — Assets	Licences, infrastructure, models, dashboards. The organisation owns something it did not own before.	Could a competitor with the same budget hold this position in six months? Usually yes.
P — Processes	The flow of work changes: approvals route differently, handoffs disappear, a queue is capped.	Would the new flow survive if the tool were replaced by a different vendor's?
C — Capability	The organisation ends up better at improving its processes than it was before.	What can it now do that it could not do before — *that does not depend on the new system?*

Training people to operate the new system is not capability. It is part of delivering the asset, and it ends when the people who were trained leave. The C test is deliberately hard to pass, because most honest answers are A, or A and P, and recording that truthfully is what makes the appraisal useful.

The productivity paradox, and why it is not about technology

Research on information technology investment has repeatedly found the same shape: the returns to technology appear where complementary organisational practices exist, and are close to nil where they do not. The pattern survived decades of changing technology because it was never about the technology.

It is the same complementarity argument from this morning, in a different literature. A practice adopted without its complements imposes its costs and withholds its returns — and a platform is a practice that arrives with an invoice attached.

IN PROJECT TERMS

THE IDEA A platform's benefit is not proportional to its adoption, and every fact needs exactly one home.

IN A PROJECT A tool used as the working record by 40% of the people it was bought for does not deliver 40% of a single-source benefit; it delivers close to none, because the other 60% means every figure still has to be reconciled. And a schedule date, a risk or a cost figure that lives in two systems — the PPM platform and the spreadsheet the team actually uses — imposes a permanent reconciliation tax, and turns the dashboard into a negotiation.

ON MONDAY List the five facts leadership asks about most. For each, write where the authoritative version lives. If any has two homes, that is the first project, and it is not a software project. Then put change management on the same budget line as licences, because the arithmetic will punish you if it is not.

The trap that looks like speed

A platform that makes work easier to start, and does not cap what is in progress, lengthens lead time rather than shortening it. Little's Law does not stop at the screen: with throughput fixed, more in flight is more time in flight. The platform delivers exactly what it promised — frictionless initiation — and the organisation experiences it as delay.

This is worth saying plainly because it is the most common disappointment in enterprise tooling, and it is almost never diagnosed correctly. The tool is not underperforming. It is performing, and nobody capped the queue.

THREE

Philosophy eats AI

Every digital transformation now passes through artificial intelligence, and AI is the extreme case of everything above: the most expensive asset an organisation can buy, and the one whose return depends most completely on a capability that cannot be bought alongside it.

Schrage and Kiron put the claim sharply. The binding constraint on sustainable value from AI is **not technical but philosophical**. Organisations invest heavily in algorithms, data infrastructure and talent while never articulating what their systems should value, how they should know, and what they should represent about reality. Those assumptions do not stay harmless because they were never written down. They are executed, at scale, continuously.

Three pillars

Each governs a different question, and each has an exact counterpart in what these two sessions have already established — which is the reason to treat them as operations content rather than as philosophy imported for decoration.

Teleology — what the system is told to want

The objectives and the hierarchy among them. It is the most critical of the three and the most frequently skipped, because a single objective is easy to state and a *ranking* of conflicting objectives requires a decision nobody wants to put in writing.

When a system optimises the wrong metric, or when objectives conflict with no explicit hierarchy, it produces outcomes that are technically correct and strategically destructive. A demand forecast optimised to minimise inventory will systematically underestimate seasonal peaks. A credit approval system optimised to reduce default will systematically exclude profitable segments.

> WHERE YOU HAVE MET THIS
>
> Order winners and qualifiers. A system optimising the wrong metric is a system whose purposes were never ranked — exactly the organisation that pours effort into a qualifier above the threshold and wonders why preference did not follow.

Epistemology — what the system may count as knowing

How it acquires knowledge, how it validates it, and how it expresses certainty. Systems that do not separate correlation from causation, that do not quantify uncertainty, or that treat historical patterns as permanent truths are epistemologically fragile.

The practical consequence is decisions taken with high statistical confidence in conditions that no longer resemble the ones that generated the training data. The pandemic invalidated the historical data behind practically every demand model in operation simultaneously — and the models kept reporting confidence throughout.

> **WHERE YOU HAVE MET THIS**
>
> Signalling at the source. A system that knows when it does not know, and escalates rather than answering, is an andon: the abnormality surfaces the moment it occurs instead of arriving in the next monthly report. A system that always returns an answer, with no stated confidence, is aggregated reporting with a faster clock.

Ontology — what the system is able to see

Which entities it recognises, which relations it models, which variables exist for it at all. A sourcing system that represents suppliers by price and quality alone — ignoring geopolitical exposure, sustainability and innovation capacity — has an impoverished ontology. Its recommendations will be locally optimal and systemically fragile.

Schrage and Kiron's own example is customer loyalty. Companies say they want loyal customers and then hand the system recency, frequency, monetary value, churn and NPS — proxies that were never loyalty, only the parts of it that could be counted. The system optimises the proxies, faithfully, and the organisation forgets that it ever meant anything else. Korzybski's line, which they quote, is the whole of ontology in six words: *the map is not the territory.* The question to ask of any system is what it has been given as a stand-in for the thing you actually care about.

> **WHERE YOU HAVE MET THIS**
>
> Causal ambiguity, from the benchmarking argument that follows. Representing a supplier by price and lead time alone is the same error as benchmarking the indicator instead of the practice that produces it: in both cases the representation omits the thing that actually explains the result.

Philosophical debt

Schrage and Kiron's phrase for the condition is *tacit, unarticulated philosophical principles*; the Gemini episode they analyse is "teleological chaos" — rival objectives allowed to reign with no hierarchy among them. The term used in this course for the accumulation is **philosophical debt** (Lima, 2026, Ch. 5): unarticulated assumptions about purpose, knowledge and reality, embedded in the models, compounding silently until the damage is hard to reverse. The analogy with technical debt is exact in form, and the difference is where the argument of this day lands.

Technical debt accumulates in A and P. Philosophical debt accumulates where C should be.

a shortcut in the artefact · an absence in the organisation

Technical debt is a compromise somebody made and could document. Philosophical debt *is* the absence of the documentation: nobody wrote down the hierarchy of purposes, so nobody can audit it, so nobody can improve it. An organisation that cannot say what its system is optimising for has no object on which to run an improvement routine. That is a deficit in capability, arriving dressed as a purchase of assets.

> **THE FAILURE THESE THREE PREDICT**
>
> A sourcing system optimised for cost and delivery which, two years on, has systematically excluded suppliers in the regions the firm's own strategy prioritised. The fault is not in the model. The correction is not technical either — adding a column does not resolve it. The hierarchy of objectives was never articulated, so the trade-off was decided by default; and without articulating it, the technical fix simply creates a new debt in the opposite direction.

The article's action plan, read against the instrument

The MIT SMR *Action Pack* for the article — the authors' own summary deck — restates the argument as four concepts and a four-step action plan. These notes take a deliberate cut of it, and it is worth being explicit about what was taken and what was left, so that anyone who reads the original is not surprised by the difference.

In the article and its action pack	Where it lands in the Digital Tool Appraisal	What the appraisal does with it
Tacit principles — companies deploy AI with unarticulated philosophical principles rather than deliberate frameworks	Section 4 — the philosophical debt audit, and the "not documented" answer it is built to surface	Taken whole. It is the premise of the audit.
Teleological architecture — navigate purpose hierarchies; resolve competing priorities with principled trade-offs	Section 4, teleology: what the system is told to want, and what it does when objectives conflict	Taken, narrowed to the one question a steering committee can answer: is the hierarchy written down?
Epistemological agency — metacognitive awareness of what the system knows, does not know, and how reliable its knowledge is	Section 4, epistemology: does it know when it does not know, and escalate? Read with condition 2 of the chain	Taken as the andon test. The article's further steps (self-directed learning, hypothesis generation) are not appraised.

Ontological understanding — causal architecture and systems thinking; self-understanding within human–AI collaboration	Section 4, ontology: what the system is able to see; read with conditions 1 and 3	Taken as the proxy question — what stands in for the thing you care about. Causal modelling of the system itself is left out.
Models as talent — treat and train models as high-potential talent; AI talent reviews; KPIs for model improvement	Not in the instrument	Left out. It is a programme for developing the model; the appraisal asks whether the organisation can use what the model already does. The six-condition chain covers the measurement side differently — by outcome, not by model performance.
Ethical agency — autonomous moral reasoning beyond rule-following	Not in the instrument	Left out, deliberately. It is a question about the system's own reasoning; the appraisal stays on what the organisation articulated and measured. The article notes that ethics is only a small part of philosophy's effect on AI, and the cut here follows that remark.

The term *philosophical debt* is not in the article or the pack; it is this course's name for the accumulation the authors describe. The action pack itself is published by MIT SMR and is not reproduced here; it is available from the publisher, and the article is on its site.

From predictive to agentic, and why the stakes change

Most AI in operations today is predictive: the system analyses or recommends, and a human decides and acts. Agentic systems formulate plans and act in pursuit of objectives without a human in each step. Along all three pillars, what was documentation becomes a control surface — the hierarchy of purposes stops being a statement of intent and becomes the place where the boundary of autonomy is actually set.

> THE QUESTION FOR THE NEXT STEERING COMMITTEE
>
> Technical safeguards — value limits, audit logs, anomaly alerts — are necessary and not sufficient. Before granting any system autonomy, three philosophical ones. **Teleology:** is the hierarchy of purposes documented, including what the system should do when cost, quality and sustainability conflict? **Epistemology:** does it know when it does not know, and escalate below a confidence threshold? **Ontology:** can it recognise a supplier or a context it has never seen, or does it only optimise inside the world it was trained on? Without those three, autonomy is risk wearing the costume of efficiency.

FOUR

From a working model to an organisational result

Part Three argued that where a tool contains AI, the binding constraint is conceptual rather than technical. That is an argument. What follows is a structure for testing it on a specific implementation — and it comes from the studies that installed AI in a real organisation, changed something, and measured what happened afterwards.

> **WHERE THE FRAMEWORK COMES FROM**
>
> It is drawn from a systematic review of AI implementations conducted under PRISMA 2020 (Lima, manuscript in preparation). **The evidence base is hospital implementations**, and the conditions are stated here in general organisational terms — an extension of the framework, not a finding of the review. Use it as a structure for appraisal, not as a citable result.

Two findings frame it. Of 1,271 records screened, only **37** had both an implementation and a comparison: roughly one for every eight studies that end at a model or a simulation. And among those that produced no effect, predictive performance stayed high wherever it was measured. **Model accuracy does not separate success from failure.** Something else does.

Six conditions, each necessary, none sufficient

Organisational value is the end of a chain. A break at any link stops the effect, however good the preceding links were. The two middle conditions are not properties of the tool at all: they are capabilities of the organisation that bought it.



- Conditions of the tool, and of the result
- The two organisational capabilities — this is C, made measurable

	Condition	The question to ask	What it looks like when it breaks
0	**Evidence base**	Has this class of application produced a measured result in a real organisation — not in a model, not in a simulation?	Modelling studies and vendor material, with no measured implementation anywhere.
1	**Technical capability in operation**	Does the model still perform on our data, at the threshold actually in use?	Pre-deployment performance not reproduced in routine use.
2	**Workflow integration**	Does the output reach whoever decides, while the decision is still open?	Two thirds of the staff never open it; or it arrives after the decision was taken.
3	**Translation into action**	When it arrives, is there an owner able to act — with the resources that action requires?	Cases correctly identified and nobody able to act: people never contacted, actions blocked on resources.
4	**Organisational outcome**	Which management indicator should move, from which source, against which baseline?	Benefit stated in general terms, or confirmed only by what users say they felt.
5	**Net value**	What does the gain cost, and who bears it?	Gains offset by work displaced to other people or other hours.

Used **before** adoption, the chain is a checklist of conditions the organisation itself controls. Used **after**, it locates the link where the implementation failed — a different and more useful question than whether the model was accurate. Conditions 2 and 3 are where the reviewed evidence says value is most often lost, and neither is something a vendor can fix.

> TWO BRIDGES, BRIEFLY
>
> **To Part Three.** The pillars ask what nobody *articulated*; the chain asks what nobody *measured*. Teleology surfaces at conditions 4 and 5, epistemology at condition 2, ontology at 1 and 3.
>
> **To project practice.** Condition 4 is benefits realisation management, which Serra and Kunc (2015) associate with project success and with the execution of strategy rather than with the delivery of outputs. Conditions 2 and 3 are almost never in a digital project's scope: the system is delivered, and the decision it enters and the capacity to act on it are left to change management after go-live.

Adopting AI is a process-design decision, not a technology purchase.

conditions 2 and 3 are the organisation's, not the vendor's

FIVE

Benchmarking: Decoding the rules-in-use

Benchmarking is conventionally practiced as corporate espionage or tourism: visiting high-performing firms, photographing their tools, buying their software, and pasting their org charts. That is why it fails.

The method, before the critique

Benchmarking has an established methodology, and the criticism that follows in this part is a criticism of how it is practised rather than of the method itself. Camp's formulation, from the Xerox programme that made the practice general, runs in four movements.

	Phase	What it requires	Where it is usually skipped
1	**Plan**	Decide what to benchmark — a process, not a department — identify a comparator, and settle how the measure will be defined before anybody goes looking.	The comparator is chosen for prestige rather than for comparability.
2	**Collect**	Gather data on both sides using the same definition, and record the conditions under which each figure was produced.	Their figure is taken from a published source; ours is calculated differently.
3	**Analyse**	Establish the size of the gap and, far more importantly, the practice that produces it.	The gap is quantified and the practice is never identified. This is the fatal one.
4	**Adapt**	Set targets, change the practice, embed it, and re-measure.	Targets are set from the gap with no change to the practice that would close it.

Choosing the comparator

Four kinds, in increasing order of difficulty and of payoff.

- **Internal** — another unit, plant or region in the same organisation. Easiest access, most comparable conditions, smallest ceiling: it can only take you to the best you already are.
- **Competitive** — a direct rival. The most wanted and the least available, and the one most likely to converge the industry, for the reasons set out below.
- **Functional** — an organisation that performs the same function in a different industry.

- **Generic** — a process that is structurally analogous in a wholly unrelated setting. Xerox famously fixed its warehouse by studying a catalogue retailer rather than a competitor, and the distance is what made the practice visible instead of invisible.

Making the numbers comparable, and using international comparators

Most benchmarking failures are arithmetic before they are strategic. Three corrections do most of the work: **define the boundaries** of the process identically on both sides — most on-time-delivery disputes are disputes about when the clock starts; **normalise for scale and mix**, since cost per unit across different volumes and product mixes is not one measure but two; and **record the conditions**, because a figure achieved under different regulation, labour markets or demand variability is not a target, it is a datum.

International comparators — standards bodies, professional-association survey data, published maturity studies — are useful for exactly one thing: establishing whether a gap exists at all, and roughly how large. They cannot tell you what produces it, because they report distributions rather than practices. Treating a percentile from a survey as an improvement objective is the error in phase 3 above, formalised and given a chart.

Why the copy-paste approach guarantees mediocrity

Returning to the NUMMI puzzle from Session 1: General Motors took tours of NUMMI for twenty years. They photographed the kanban boards, copied the andon lights, and replicated the physical workstations in their other plants. The other plants remained unproductive and combative.

Why? Because GM copied **Assets (A)** and documented **Processes (P)**. They could not observe, let alone import, the tacit **Capabilities (C)** and the *Rules-in-Use*.

$$V_{rival} = C_{weak} \times (A_{bought} + P_{copied}) \approx 0$$

The mathematical explanation of failed benchmarking

Why best practice resists transfer: the evidence

NUMMI is not an anecdote about one stubborn company. It is the most visible instance of a finding that strategy and operations research reached independently, and four results carry it.

It fails even inside the same firm. Szulanski (1996) studied transfer between units of the *same* organisation — no competitive barrier, shared incentives, cooperative intent — and found it stalled anyway. The obstacles were not motivational, which is what managers assume, but obstacles of knowledge: ambiguity about why the practice works, and limited capacity in the receiving unit to absorb it (Cohen & Levinthal, 1990). He called it *internal stickiness*. If a practice will not travel reliably between two plants of one company, assuming it travels from a competitor through a site visit is not conservative. It is heroic.

Observation is not understanding. Organisational knowledge lives in routines that are largely tacit and that no document fully specifies (Nelson & Winter, 1982; Polanyi, 1966). Under **causal ambiguity**, even the firm that owns a superior practice may not know which elements produce the result (Lippman & Rumelt, 1982) — so it cannot reliably teach it, and a rival cannot reliably copy it. GM's visitors were not looking at the wrong things. They were looking at the only things that could be looked at.

Partial copies perform worse than no copy. Where a strategy is a system of interdependent choices, a rival who gets most elements right and a few wrong typically performs worse than one who copies nothing, because the mismatched elements work against each other (Rivkin, 2000). This is the complementarity argument of the morning in formal dress, and the precise reason GM's other plants — kanban boards installed, andon lights fitted, the rest absent — got worse rather than better.

Capability is accumulated, not purchased. Building in one year what a rival built in five costs disproportionately more and often cannot be done at any price (Dierickx & Cool, 1989). Winter (2003) then draws the distinction that organises the session: an *operational* capability is how the firm earns its living today, a *dynamic* one is its capacity to change the operational ones. In the language of the morning, **P is the operational capability and C is the dynamic one.**

Mechanism	What it predicts	What it changes in your study
Internal stickiness Szulanski (1996)	Transfer fails for knowledge reasons, not motivational ones	Budget for absorption, not only for acquisition. Who here can receive this, and what must they already know?
Causal ambiguity Lippman & Rumelt (1982)	Even the source may not know which elements matter	Do not ask the reference what it does. Ask what it stopped doing, and what it would never give up.
Interdependence Rivkin (2000)	Partial imitation underperforms no imitation	Identify the bundle, not the practice. If you cannot adopt the set, adopting one element may cost you.
Time compression Dierickx & Cool (1989)	Accumulated capability cannot be bought quickly at any price	Ask how long the reference took. Then plan for that order of magnitude, or change the objective.

> **THE RULE THIS GIVES YOU**
>
> **Do not benchmark the number. Benchmark the routine that produces it — then ask what your organisation would have to accumulate in order to run that routine.** The first question has an answer you can put on a slide. The second has an answer you can act on.

THE STRATEGIC CAVEAT

There is a second-order risk the literature above does not address, and Porter (1996) does. Benchmarking is by construction an *imitative* activity: it moves the organisation towards a frontier others already occupy. Practised well across an entire industry, it converges competitors onto the same operating model, and competition collapses into price.

The morning gave this its arithmetic. Competitive advantage was written as a difference — $VC = (WTP_A - Cost_A) - (WTP_B - Cost_B)$ — and a successful imitation is exactly the event that pulls the second bracket up towards the first. Benchmarking is therefore the one improvement activity whose logical endpoint, when both sides practise it well, is VC equal to zero for everyone in the industry.

That is not an argument against benchmarking. It is an argument for being explicit about which of two things you are doing — closing a gap on a qualifier, where convergence is precisely the goal, or building an order winner, where imitation is the wrong instrument entirely.

Spear & Bowen's Four Rules applied to Digital Benchmarking

The theory says what to look for. Spear and Bowen (1999) say where to look. Their four rules are the decoding key — the tacit operating rules a site visit will never surface unless the visitor arrives knowing they exist.

Rule	What naive benchmarking sees	What rigorous benchmarking decodes
1. Standardized Work	"They have standard operating procedures and software forms."	Work is specified to the level of content, sequence, timing, and direct observable outcome. Any variance is treated as a testable scientific problem.
2. Direct Connections	"They use a central ticketing platform or Slack channel."	Every customer-supplier connection within the organization is binary, direct, and unambiguous, with zero intermediate translation or ambiguous handoffs.
3. Simple Pathways	"They have automated end-to-end routing in their ERP."	Every product, service, or safety signal travels along a pre-specified, single pathway. No work loops back or diverts unless explicitly designed.
4. Scientific Method	"They run monthly retrospective meetings."	Any improvement must be structured as a refutable hypothesis, mentored by a senior leader, and executed at the lowest possible organizational level.

SIX

The transformation vector — how this is actually run

Everything so far has been diagnosis: what the money bought, where the chain breaks, why the copy underperforms the original. Diagnosis ends on a question no business case answers. If the path cannot be known in advance, what replaces the plan?

Two instruments, drawn from Chapter III of *Excelência Operacional*, where Cynefin and Toyota Kata are read as one system. The order matters: Cynefin says which method the problem admits, and Kata is the routine that executes it.

The domain decides the method

Cynefin — Welsh for *habitat* — is a sense-making instrument. Its premise is that different contexts require fundamentally different styles of decision, and that the characteristic error of leadership is not a bad decision but a decision made with the method of the wrong domain.

Domain	Nature of the problem	Approach	The trap
Simple	Predictable; cause and effect obvious to everyone	Sense → Categorise → Respond	Rigidity; blindness to a change of context
Complicated	Requires expertise and analysis; several right answers exist	Sense → Analyse → Respond	Paralysis by analysis; dependence on specialists
Complex	Cause and effect visible only in hindsight. **Innovation, culture and digital transformation live here**	Probe → Sense → Respond. Safe-to-fail experiments	Wanting control before acting; planning what can only be discovered
Chaotic	No perceptible causal relation; speed over precision	Act → Sense → Respond	Analysis during a crisis
Confusion	Not knowing which domain you are in. The most dangerous state	Break the problem up and reclassify each part	Applying a preferred style with no diagnosis at all

Most leaders are trained for the Complicated domain, and that training serves daily operational problems well. The difficulty is that the highest-stakes problems — digital transformation among them — are inherently Complex, and applying Complicated methods there is not merely ineffective: a detailed plan manufactures a sense of control that delays experimentation and penalises learning.

Netflix and Blockbuster, 2007

In 2007 there was no best practice available to establish whether consumers would pay for streaming. Netflix operated the problem as Complex: it launched streaming as a free benefit for DVD subscribers (**probe**), observed adoption (**sense**), and only then pivoted the business model (**respond**). Blockbuster treated the same question as Simple and applied historical best practice.

The same market, the same moment, two opposite readings of the domain. One of the two firms was gone by 2010.

WHY THIS INDICTS THE PREVIOUS PART

Benchmarking is a Simple-and-Complicated instrument: it presupposes that a knowable right answer exists somewhere and that the task is to locate and transfer it. Pointed at a Complex problem it does something worse than fail — it imports a confident answer to somebody else's question, and the confidence is what does the damage. That is the mechanism behind Part Five's finding that a partial imitation can underperform not imitating at all.

Kata: the routine that makes probe–sense–respond repeatable

Probe → Sense → Respond is a logic, not a practice. Without a routine it is indistinguishable from improvisation and decays the moment the urgency passes. Kata — in the martial arts, a pattern rehearsed until it becomes reflex — is a routine of scientific thinking applied daily. It does not prescribe what to do. **It prescribes how to think.**

	Step	What it requires	Horizon
1	**Direction or challenge**	Aspirational enough to orient, concrete enough to guide daily decisions.	1–3 years
2	**Grasp the current condition**	Go and observe the process as it actually is, and share what is found so the next cycle does not restart from zero.	Continuous
3	**Set the target condition**	A measurable future state, neither trivial nor impossible — in the zone of productive learning.	Up to 3 months
4	**Run experiments**	Plan–do–check–act, daily. The path is deliberately obscure; were it clear, there would be nothing to learn.	Weekly / daily

THE COACHING ROUTINE — FIVE QUESTIONS

1. What is the target condition? **2.** What is the current condition? **3.** What obstacles are in the way, and which one are you addressing now? **4.** What is your next step, and what do you expect to happen? **5.** When can we see what you learned from that step?

The point is not to solve the problem. It is to produce people who solve problems — the only form in which capability survives the person who had it.

This also explains the commonest failure of Lean programmes: the tools are adopted, and the improvement does not survive six months. The tools were the visible content; the routine was the form that sustained them. And the claim Kata makes on capability is the one that carries into the close of the day — organisations have excellent principles and poor routines, and **it is the routines that determine the real culture, not the values on the wall.**

THE DILEMMA THIS PART EXISTS TO ANSWER

A competitor implements generative AI and cuts cost by 30%. What do you do?

1 · Cynefin. Is this Simple — copy the technology — or Complex — discover how AI could create value specific to your customer? **2 · Lean.** Is human contact a driver of willingness to pay in your segment? If it is, automating it cuts cost and destroys revenue at the same time. **3 · Kata.** Set a ninety-day target condition and run controlled experiments towards it, rather than approving a programme whose benefits nobody can yet state.

The reflex to copy usually reproduces operational effectiveness without producing differentiated advantage — which, in the morning's notation, leaves VC exactly where it was.

IN PROJECT TERMS

THE IDEA A gate asks for a plan. A Complex problem can only honestly supply a hypothesis.

IN A PROJECT So the plan gets written anyway, and the organisation then manages the plan instead of the problem — holding people to dates that were never knowledge, only politeness. The reconciliation is not to abandon governance but to change what an early gate receives: a target condition and the evidence from the last experiment, not a schedule and a projection.

ON MONDAY Rewrite one gate deliverable in this form: "By day 90, *this measure* will read *this value*; here is what the last four weeks showed." If the sponsor will not accept that in place of a plan, the problem is being governed as Complicated, and the next quarter will prove it.

SEVEN

Structured activity — the Digital Tool Appraisal

The morning asked whether a project was worth funding. This afternoon asks a harder question about the same project: of the digital tooling it depends on, what is actually being bought — and what is being left undone?

> **THE INSTRUMENT**
>
> The appraisal is a separate page, fillable and exportable in the same way as the morning's canvas: **Digital Tool Appraisal**. Open it now. In the room, groups complete sections 1 to 5 — the tool, the VCAP triage, the six conditions, the AI audit where it applies, and the verdict — in about twenty minutes. Sections 6 and 7, the reference and the six-month roadmap, are the module's follow-up assignment and are better done with a week of thought. The page carries its own instructions; what follows is why it asks what it asks.

Four layers, and why they are one instrument

The appraisal does not inventory features. A list of whether the tool has a poka-yoke, how many systems people toggle between and whether an andon exists is an audit, and an audit is work for one person with a checklist — not for four senior managers with twenty minutes. What four senior managers can do that a checklist cannot is **judge**, and each layer forces a judgement that is uncomfortable to make honestly.

Layer	The judgement it forces	Where it comes from
VCAP triage	Does this change what we own, how work flows, or what we are able to improve? Most honest answers land on the first two.	This morning's identity, V = C × (A + P)
Philosophical debt	Only where the tool contains AI: has anyone written down what it should want, what it may count as knowing, and what it is able to see?	Part Three — Schrage & Kiron's three pillars
The six conditions	For any tool: at which link does the chain from a working tool to a result currently break, or go unmeasured? This is the maturity read.	Part Four — the six-condition chain
The reference	What does somebody already do that we do not — the routine, not the number — and what would have to be true here for it to work?	Part Five — the method, then internal stickiness, causal ambiguity and time compression

Maturity and roadmap	Which capability gaps most constrain performance, and in what order can they credibly be closed over six months?	The six conditions read across, plus strategic value against feasibility

The fifth layer is where the instrument stops being about one tool. Read across, the six conditions are a maturity statement about the organisation — which conditions it can satisfy and demonstrate, which it satisfies without evidence, and which it has never observed. The instrument deliberately produces no composite maturity score: a single figure would conceal the only thing worth knowing, which is *where* the chain stops. The roadmap then sequences the three largest gaps by strategic value against feasibility, which is the output the module's assignment asks for.

A tool can pass the pillars and fail the chain — a clearly stated purpose whose output reaches nobody in time — and it can pass the chain and fail the pillars, which is the more dangerous case: an implementation that demonstrably moves an indicator, with nobody able to say which other indicator is paying for it.

> **WHERE THE VERDICT LEADS**
>
> A verdict is not a score, and the instrument does not produce one. It produces a located gap — and Part Six says what to do with it. If the gap is at condition 2 or 3, the problem is Complex: the integration into somebody's decision and the capacity to act cannot be specified in advance, so the next move is a target condition at ninety days and an experiment, not a requirements document. If the gap is a missing measure, the next move is simply to measure it before spending anything further.

What to expect

"Not documented" will be the honest answer

Most groups find that nobody can say what the system optimises for when objectives conflict. That is the finding, not a failure of the exercise — and it is cheap to fix before deployment and expensive afterwards, once the organisation has two years of decisions made under the implicit hierarchy.

Most tools will land on A, or A and P

That is a correctly identified purchase, not a weakness. What it means is that a competitor with the same budget can hold the same position — so the business case should be argued on that basis rather than on advantage.

Expect, too, that the six conditions will be answered *not measured* more often than *no*. That is the characteristic result and it is worth saying out loud: an organisation that cannot say whether the output

reached the decision in time, or whether the recommended action was ever carried out, does not have a failing implementation — it has an unobserved one, which is a different and more tractable problem.

Keep the exported appraisal. Together with the morning's Value Case it is the material the module's group Execution Optimization Challenge is built from — the coordination will set out the brief.

EIGHT

Building the multiplier

The day has been an argument by elimination. Assets can be bought. Processes can be documented and taught. Benchmarking transfers the second and purchases the first, and neither moves the term that multiplies them. One question is left, and it is the only one with a practical answer: **how is C actually built?**

Capability is a stock, not a state

Dierickx and Cool's distinction, from Part Five, organises everything here: flows adjust instantly, stocks do not. Treating the capability to improve processes as a stock means asking three separate questions — what flows in, what flows out, and what conditions regulate both.

	Mechanism	Managerial consequence
Inflow	Deliberate improvement activity that produces causal understanding and validated action together	Volume of improvement projects is not the measure. Projects that produce only one of the two do not accumulate anything.
Condition	Psychological safety and the credibility of management's commitments	Where surfacing a problem is costly, the measured performance improves and the real performance does not.
Outflow	Turnover and disuse	An organisation that was good at improving five years ago may no longer be, and nothing on its balance sheet will say so.

What the inflow actually requires

The morning established the finding: across improvement projects in a single factory, only about a quarter accelerated the reduction of waste — the ones producing *know-why* and *know-how* simultaneously (Lapré et al., 2000). The rest slowed it or had no effect. Three conditions follow, and Part Six has already named the routine that satisfies the first.

- **Guided experimentation at the point of execution** — improvement run as a hypothesis, tested by the people doing the work, under a more experienced coach. This is Spear and Bowen's fourth rule read as an accumulation mechanism, and it is the improvement kata operationalised.
- **Team learning** — capability lives in coordinated routines, not in individuals. A team that cannot discuss a failure cannot refine a routine, which is why psychological safety is a precondition rather than a nicety.

- **Credibility** — sustained effort depends on management's promises being believed, above all the promise not to convert improvement gains into redundancies. People do not improve away their own jobs, and they are right not to. Break it once and the inflow stops.

> **IN PROJECT TERMS**
>
> **THE IDEA** The inflow and the outflow both have project equivalents that most organisations already own and few use as intended.
>
> **IN A PROJECT** The retrospective is guided experimentation — if its output changes the standard, and nothing if it does not. The lessons-learned repository is an attempt to arrest the outflow, and it fails for exactly the reasons Part Five gives: it stores what can be written down, which is the part that was never the constraint. The practical test for a PMO: after the last three projects closed, what can this organisation now do that it could not do before — that does not depend on the people who happened to be on those teams?
>
> **ON MONDAY** Open the last three retrospectives and look for one line in a template, a checklist or a gate criterion that changed because of them. If there is none, the retrospectives produced know-why without know-how, and the stock did not move.

Design is not the same as practice

Everything above describes something an organisation can *design*: which capability it sets out to build, which assets it buys, which processes it writes down, which improvement routine it adopts. All of that is specifiable, auditable, and — the whole lesson of the afternoon — visible to a competitor. Design is necessary and it has never been sufficient, because the same design produces different results in different organisations. That gap has a name, and a form:

$$\text{Operational Excellence} = \text{Design(VCAP)} \times \text{Culture}$$

what the organisation specified · multiplied by whether it is actually lived

The formulation above is the author's adaptation. The structure it borrows — a designed solution *multiplied* by its acceptance in practice rather than added to it — is long-standing in the literature on organisational change; the reading of the first term as a VCAP configuration is this course's.

Design(VCAP) is the deliberate configuration: the chosen capability, the assets acquired, the processes written. It is the blueprint, and it is the part that can be presented, bought, certified and copied.

Culture is the substrate the whole configuration sits on — not a fourth term beside A, P and C, but the condition under which any of them delivers what it was designed to deliver. It is what the organisation does when the auditor has left: whether the standard is followed when it is inconvenient, whether the andon is

pulled when pulling it is embarrassing, whether the improvement routine survives a quarter in which the numbers are bad.

> **WHERE C ENDS AND CULTURE BEGINS**
>
> **C is inside the design; Culture is whether the design is real.** C is a capability the organisation chose and can describe — "we are able to improve this class of process repeatedly". Culture is not described, it is observed, and it governs every term at once: an asset used as instructed and an asset worked around, a documented process followed and the same process bypassed, a declared improvement capability exercised and one that exists only in the handbook. That is why it multiplies the expression rather than sitting inside it.

The form is multiplicative for the same reason VCAP is. An addition would say that a strong design partially compensates for an organisation that does not live it — that excellent processes plus indifferent practice yield moderate excellence. They do not. They yield binders. At the limit the statement is blunt: where Culture is zero, Operational Excellence is zero, whatever was designed and whatever was spent.

> **WHAT THIS FORMULATION IS, AND IS NOT**
>
> It is a didactic statement of a relation, not an instrument. Culture has no unit here and no score, and any attempt to supply one would be inventing precision the concept does not carry. The purpose of writing it as a product is to forbid an addition — to make it impossible to argue that a good design and a weak practice average out to something acceptable. Used as a reminder it is sound; used as a measurement it is not.

Where the day ends

$$V = C \times (A + P)$$

A is bought · P is copied · C is the only term you have to build

This morning opened on a plant in Fremont where the assets and the people stayed the same and the result inverted. The afternoon has given that observation its mechanism: what changed was a stock that no visitor could photograph, no contract could transfer, and no platform could install — and which, left alone, would have drained away again within a few years. Read through the second formulation, Fremont is the cleanest experiment available: the design changed, and so did the substrate it ran on, and only the second explains why the same plant and the same workforce produced a different outcome.

NINE

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